Lab6

Course: ARTI 404 – Image Processing

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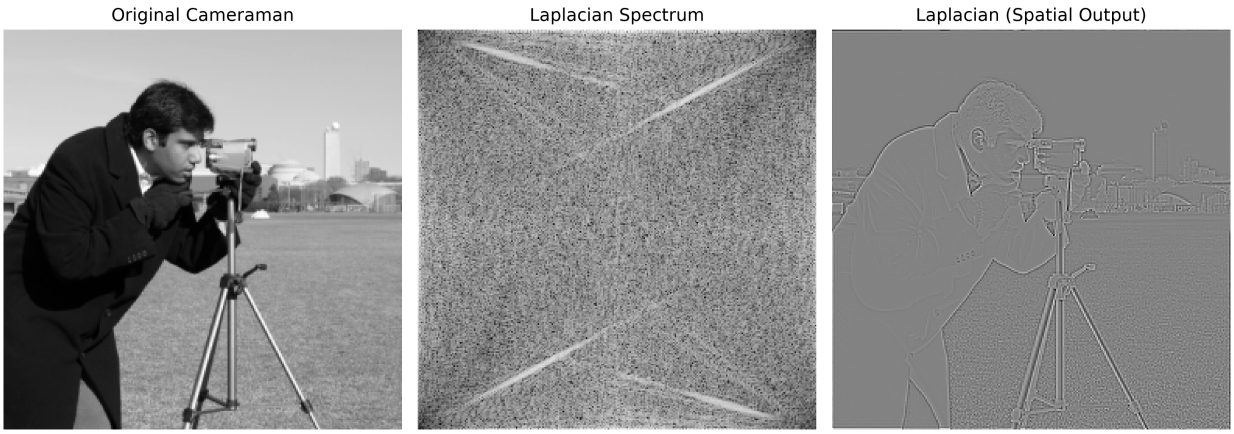
Student ID: 2240006060

Procedural Task#1

Apply a Laplacian filter in the frequency domain to the cameraman image. In the frequency domain, the 2D Laplacian operator corresponds to multiplying the Fourier transform F(u, v) by $-4\pi^2(u^2 + v^2)$. Resize the image to 256x256, compute its FFT, multiply it by the Laplacian filter, apply the inverse FFT, and show the original image, the Laplacian spectrum, and the spatial output side by side.

```
# Load cameraman image (already grayscale) and resize for faster FFT
image= data.camera()
image= resize(image, (256, 256), anti_aliasing= True)
# Save original image for repository submission
plt.imsave("images/cameraman.png", image, cmap= "gray")
# Get image dimensions
M, N = image.shape
# Generate frequency grid
u= fftfreq(M).reshape(-1, 1)
v= fftfreq(N).reshape(1, -1)
# Compute Laplacian filter in frequency domain
laplacian_filter= -4 * (np.pi ** 2) * (u**2 + v**2)
# Apply 2D FFT to the image
F= fft2(image)
# Apply Laplacian filter in frequency domain
F_lap= F * laplacian_filter
# Inverse FFT to obtain spatial result
laplacian_image= ifft2(F_lap).real
# Plot results
plt.figure(figsize= (12, 5))
plt.subplot(1, 3, 1)
plt.imshow(image, cmap= "gray")
plt.title("Original Cameraman")
plt.axis("off")
plt.subplot(1, 3, 2)
plt.imshow(np.log(1 + np.abs(F_lap)), cmap= "gray")
plt.title("Laplacian Spectrum")
plt.axis("off")
plt.subplot(1, 3, 3)
```

```
plt.imshow(laplacian_image, cmap= "gray")
plt.title("Laplacian (Spatial Output)")
plt.axis("off")
plt.tight_layout()
plt.savefig("images/task1_laplacian_results.png", dpi= 300, bbox_inches= "tight")
plt.show()
```



Assessment Task#1

Apply the 3x3 Sobel X and Sobel Y filters to the cameraman image in the frequency domain using the Convolution Theorem: g(x, y) = f(x, y) * h(x, y) ⟺ G(u, v) = F(u, v) · H(u, v). Zero-pad each Sobel kernel to 256x256, center it, and shift it with ifftshift so the kernel origin (0, 0) aligns with the top-left corner before computing its 2D FFT. Show the original image, the filtered spectra, the Sobel X and Y outputs, and the combined gradient magnitude.

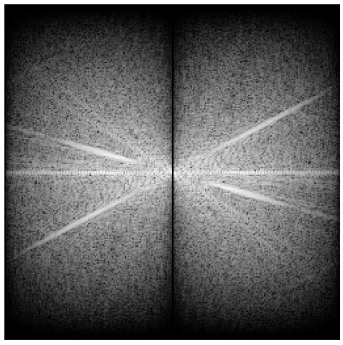
```
# Define 3x3 Sobel filters for vertical (X) and horizontal (Y) edge detection
sobel_x= np.array([[-1, 0, 1],
                   [-2, 0, 2],
                   [-1, 0, 1]])
sobel_y= np.array([[-1, -2, -1],
                   [ 0,  0,  0],
                   [ 1,  2,  1]])
# Proper padding and centering of kernel in spatial domain
def center_embed_kernel(kernel, shape):
    padded= np.zeros(shape)
    kh, kw = kernel.shape
    ph, pw = shape
    cy, cx = ph // 2, pw // 2 # Find center coordinates
    padded[cy - kh//2:cy - kh//2 + kh, cx - kw//2:cx - kw//2 + kw]= kernel # Insert
kernel centered at the image center
    return fft2(ifftshift(padded))  # Shift before FFT to place (0,0) at top-left
# Compute 2D FFT of the image
F= fft2(image)
# Compute FFT of centered Sobel filters
```

```
H_sobel_x= center_embed_kernel(sobel_x, image.shape)
H_sobel_y= center_embed_kernel(sobel_y, image.shape)
# Multiply in the frequency domain
F_sobel_x= F * H_sobel_x
F_sobel_y= F * H_sobel_y
# Compute Inverse FFT to get spatial domain filtered images
sobel_x_spatial= ifft2(F_sobel_x).real
sobel_y_spatial= ifft2(F_sobel_y).real
# Compute Combined Sobel Gradient Magnitude
sobel_magnitude= np.hypot(sobel_x_spatial, sobel_y_spatial)
# Plot Frequency Spectra and Spatial Outputs
plt.figure(figsize= (15, 8))
# Row 1: Original & Frequency Spectra
plt.subplot(2, 3, 1)
plt.imshow(image, cmap= "gray")
plt.title("Original Cameraman")
plt.axis("off")
plt.subplot(2, 3, 2)
plt.imshow(np.log(1 + np.abs(fftshift(F_sobel_x))), cmap= "gray")
plt.title("Sobel X Filtered Spectrum (Centered)")
plt.axis("off")
plt.subplot(2, 3, 3)
plt.imshow(np.log(1 + np.abs(fftshift(F_sobel_y))), cmap= "gray")
plt.title("Sobel Y Filtered Spectrum (Centered)")
plt.axis("off")
# Row 2: Spatial Domain Outputs
plt.subplot(2, 3, 4)
plt.imshow(sobel_x_spatial, cmap= "gray")
plt.title("Sobel X (Vertical Edges)")
plt.axis("off")
plt.subplot(2, 3, 5)
plt.imshow(sobel_y_spatial, cmap= "gray")
plt.title("Sobel Y (Horizontal Edges)")
plt.axis("off")
plt.subplot(2, 3, 6)
plt.imshow(sobel_magnitude, cmap= "gray")
plt.title("Combined Sobel Gradient Magnitude")
plt.axis("off")
plt.tight_layout()
plt.savefig("images/assessment_sobel_results.png", dpi= 300, bbox_inches= "tight")
plt.show()
```

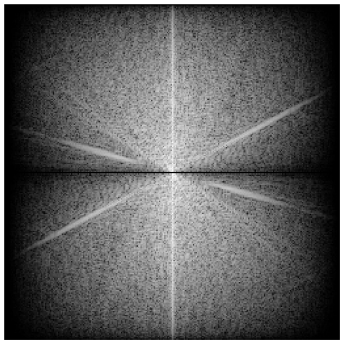
Original Cameraman



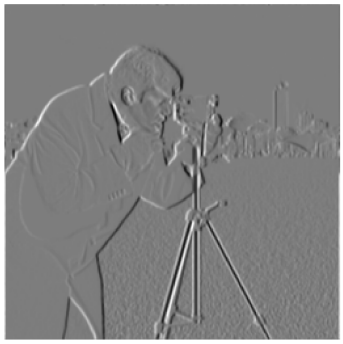
Sobel X Filtered Spectrum (Centered)



Sobel Y Filtered Spectrum (Centered)



Sobel X (Vertical Edges)



Sobel Y (Horizontal Edges)



Combined Sobel Gradient Magnitude

